WIND ENERGY ESSAY QUESTIONS

1. Explain the different types of winds?

Planetary Winds:

The winds blowing throughout the year from one latitude to another in response to latitudinal differences in air pressure are called “planetary or prevailing winds”. They involve large areas of the globe.

:Local Winds

Local winds are the ordinary winds. They are influenced by various landforms such as vegetation, hill, plains, water bodies, mountains and so on. The blow variedly and the changes are because of different temperatures and pressure regions during the night and day.

Local winds are the kind of winds that are focused as part of daily weather by the meteorological department on broadcast media such as radio and TV. The speeds of local winds range from mild to strong but just for a few hours, and they only blow over short distances. Common examples of local winds are the land and sea breezes, and valley and mountain breezes.

2. Derive the expression for maximum power that can be extracted from the wind?

We first show that for all wind turbines, wind power is proportional to wind speed cubed. Wind energy is the kinetic energy of the moving air. The kinetic energy of a mass m with the velocity v is

$$E_{kin} = ½\ m\ v^2$$

The air mass m can be determined from the air density ρ and the air volume V according to

$$m = \rho\ V$$

Then,

$$E_{kin,\ wind} = ½\ V\ \rho\ v^2$$

Power is energy divided by time. We consider a small time, Δt, in which the air particles travel a distance $s = v\ \Delta t$ to flow through. We multiply the distance with the rotor area of the wind turbine, A, resulting in a volume of

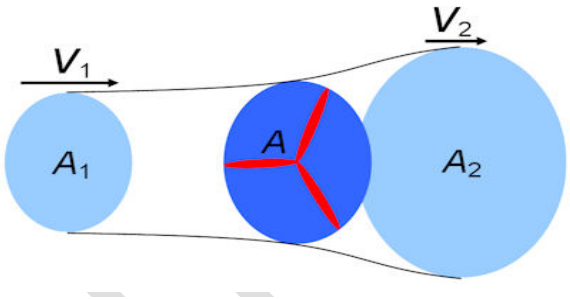
$$\Delta V = A\ v\ \Delta t$$

which drives the wind turbine for the small period of time. Then the wind power is given as

$$P_{wind} = \frac{E_{kin,\ wind}}{\Delta t} = \frac{\Delta V\ \rho\ v^2}{2\ \Delta t} = \frac{\rho\ A v^3}{2}$$

The wind power increases with the cube of the wind speed. In other words: doubling the wind speed gives eight times the wind power. Therefore, the selection of a "windy" location is very important for a wind turbine.

The effective usable wind power is less than indicated by the above equation. The wind speed behind the wind turbine can not be zero, since no air could follow. Therefore, only a part of the kinetic energy can be extracted. Consider the following picture:



The wind speed before the wind turbine is larger than after. Because the mass flow must be continuous, the area A_2 after the wind turbine is bigger than the area A_1 before. The effective power is the difference between the two wind powers:

$$P_{eff} = P_1 - P_2 = \frac{\Delta V\ \rho}{2\ \Delta t}\left(v_1^2 - v_2^2\right) = \frac{\rho\ A}{4}\left(v_1 + v_2\right)\left(v_1^2 - v_2^2\right)$$

If the difference of both speeds is zero, we have no net efficiency. If the difference is too big, the air flow through the rotor is hindered too much. The power coefficient c_p characterizes the relative drawing power:

$$c_P = \frac{P_{eff}}{P_{wind}} = \frac{\left(v_1 + v_2\right)\left(v_1^2 - v_2^2\right)}{2 v_1^3} = \frac{(1+x)\left(1-x^2\right)}{2}$$

To derive the above equation, the following was assumed: $A_1 v_1 = A_2 v_2 = A\ (v1+v2) / 2$. We designate the ratio $v2/v1$ on the right side of the equation with x. To find the value of x that gives the maximum value of C_P, we take the derivative with respect to x and set it to zero. This gives a maximum when $x = 1/3$. Maximum drawing power is then obtained for $v_2 = v_1 / 3$, and the ideal power coefficient is given by

$$c_P = \frac{P_{eff}}{P_{wind}} = \frac{16}{27} \approx 59\%$$

3. Explain the site selection for WECS?

The power available in the wind increases rapidly with the speed, hence wind energy conversion machines should be located preferable in areas where the winds are strong and persistent. Although daily winds at a given site may be highly variable, the monthly and especially annual average are remarkably constant from year to year.

The major controbution to the wind power available at a given site is actually made by winds with speeds above the average. Nevertheless, the most suitable sites for wind turbines would be found in areas where the annual average wind speeds are known to be moderately high or high.

The site choice for a single or a spatial array of WECS is an important matter when wind electrics is looked at from the systemspoint of view of aeroturbine generators feeding power into a convertional electric grid.

If the WECS sites are wrongly or poorly chosen the net wind electrics generated energy per year may be sub optimal with resulting high capital cost for the WECS apparatus, high costs for wind generated electric energy, and low Returns on Investment. Even if the WECS is to be a small generator not tied to the electric grid, the sitting must be carefully chosen if inordinately long break even times are to be avoided. Technical, Economic, Evironmental, Social and Other actors are examined before a decision is made to erect a generating plant on a specific site.

1. **High annual average wind speed:**

The speed generated by the wind mill depends on cubic values of velocity of wind, the small increases in velocity markedly affect the power in the wind. For example, Doubling the velocity, increases power by a factor of 8. It is obviously desirable to select a site for WECS with high wind velocity. Thus a high average wind velocity is the principle fundamental parameter of concern in initially appraising WESCS site. For more detailed estimate value, one would like to have the average of the velocity cubed.

2. **Availability of anemometry data:**

It is another improvement sitting factor. The aenometry data should be available over some time period at the precise spot where any proposed WECS is to be built and that this should be accomplished before a sitting decision is made.

3. **Availability of wind $V_{(t)}$ Curve at the proposed site:**

This important curve determines the maximum energy in the wind and hence is the principal initially controlling factor in predicting the electrical output and hence revenue return o the WECS machines.

It is desirable to have average wind speed 'V' such that V>=12-16 km/hr (3.5 – 4.5 m/sec) which is about the lower limit at which present large scale WECS generators 'cut in' i.e., start turning. The $V_{(t)}$ Curve also determines the reliability of the delivered WECS generator power, for if the $V_{(t)}$ curve goes to zero there be no generated power during that time.

If there are long periods of calm the WECS reliability will be lower than if the calm periods are short. In making such realiability estimates it is desirable to have measured $V_{(t)}$ Curve over about a 5 year period for the highest confidence level in the reliability estimate.

4. Wind structure at the proposed site:

The ideal case for the WECS would be a site such that the $V_{(t)}$ Curve was flat, i.e., a smooth steady wind that blows all the time; but a typical site is always less than ideal. Wind specially near the ground is turbulent and gusty, and changes rapidly in direction and in velocity. This depature from homogeneous flow is collectively referred to as "the structure of the wind".

5. Altitude of the proposed site:

It affects the air density and thus the power in the wind and hence the useful WECS electric power output. Also, as is well known, the wind tend to have higher velocities at higher altitudes. One must be carefully to distinguish altitude from height above ground. They are not the same except for a sea level WECS site.

6. Terrain and its aerodynamic:

One should know about terrain of the site to be chosen. If the WECS is to be placed near the top but not on the top of a not too blunt hill facing the prevailing wind, then it may be possible to obtain a 'speed-up' of the wind velocity over what it would otherwise be. Also the wind here may not flow horizontal making it necessary to tip the axis of the rotor so that the aeroturbine is always perpendicular to the actual wind flow.

It may be possible to make use of hills or mountains which channel the prevailing wind into a pass region, thereby obtaining higher wind power.

7. Local Ecology

If the surface is base rock it may mean lower hub height hence lower structure cost. If trees or grass or vegetation are present, all of which tend to destructure the wind, the higher hub heights will be needed resulting in larges system costs that the bare ground case.

8. Distance to road or railways:

This is another factor the system engineer must consider for heavy machinery, structure, materials, blades and other apparatus will have to be moved into any choosen WECS site.

9. Nearness of site to local centre/users:

This obvious criterion minimizes transmission line length and hence losses and cost. After applying all the previous string criteria, hopefully as one narrows the proposed WECS sites to one or two they would be relatively near to the user of the generated electric energy.

10. Nature of ground:

Ground condition should be such that the foundation for a WECS are secured. Ground surface should be stable. Erosion problem should not be there, as it could possibly later wash out the foundation of a WECS, destroying the whole system.

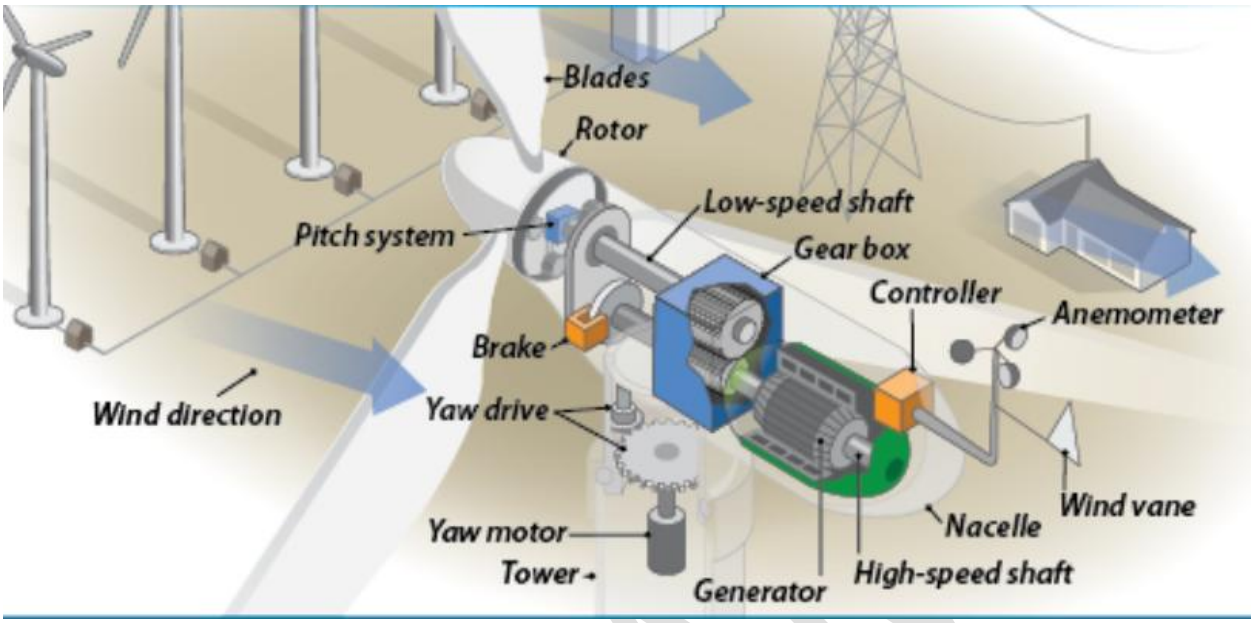
11. Favourable land cost:

Land cost should be favourable as this along with other siting costs, enters into the total WECS system cost.

12. Other conditions such as icing problem, salt spray or blowing dust should not present at the site, as they may affect aeroturbine blades or environmental is generally adverse to machinery and electrical apparatus.

4. Explain the basic components of WECS with neat sketch?

Wind turbines harness the power of the wind and use it to generate electricity. Simply stated, wind turbines work the opposite of a fan. Instead of using electricity to make wind—like a fan—wind turbines use wind to make electricity. The wind turns the blades, which in turn spins a generator to create electricity. This illustration provides a detailed view of the inside of a wind turbine, its components, and their functionality.



Anemometer:

Measures the wind speed and transmits wind speed data to the controller.

Blades:

Lifts and rotates when wind is blown over them, causing the rotor to spin. Most turbines have either two or three blades.

Brake:

Stops the rotor mechanically, electrically, or hydraulically, in emergencies.

Controller:

Starts up the machine at wind speeds of about 8 to 16 miles per hour (mph) and shuts off the machine at about 55 mph. Turbines do not operate at wind speeds above about 55 mph because they may be damaged by the high winds.

Gear box:

Connects the low-speed shaft to the high-speed shaft and increases the rotational speeds from about 30-60 rotations per minute (rpm), to about 1,000-1,800 rpm; this is the rotational speed required by most generators to produce electricity. The gear box is a costly (and heavy) part of the wind turbine and engineers are exploring "direct-drive" generators that operate at lower rotational speeds and don't need gear boxes.

Generator:

Produces 60-cycle AC electricity; it is usually an off-the-shelf induction generator.

High-speed shaft:
Drives the generator.

Low-speed shaft:
Turns the low-speed shaft at about 30-60 rpm.

Nacelle:
Sits atop the tower and contains the gear box, low- and high-speed shafts, generator, controller, and brake. Some nacelles are large enough for a helicopter to land on.

Pitch:
Turns (or pitches) blades out of the wind to control the rotor speed, and to keep the rotor from turning in winds that are too high or too low to produce electricity.

Rotor:
Blades and hub together form the rotor.

Tower:
Made from tubular steel (shown here), concrete, or steel lattice. Supports the structure of the turbine. Because wind speed increases with height, taller towers enable turbines to capture more energy and generate more electricity.

Wind direction:
Determines the design of the turbine. Upwind turbines—like the one shown here—face into the wind while downwind turbines face away.

Wind vane:
Measures wind direction and communicates with the yaw drive to orient the turbine properly with respect to the wind.

Yaw drive:
Orients upwind turbines to keep them facing the wind when the direction changes. Downwind turbines don't require a yaw drive because the wind manually blows the rotor away from it.

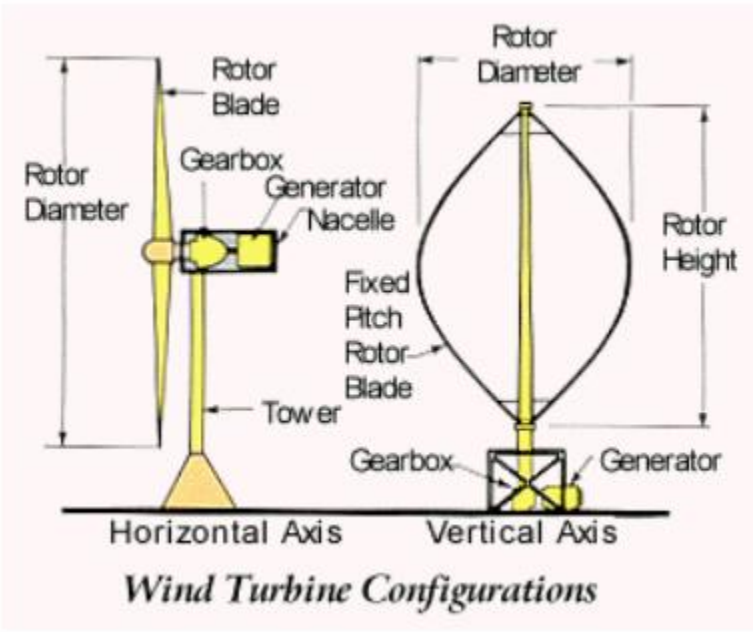
Yaw motor:
Powers the yaw drive.

5. Write the classification of WECS?

It is a mechanical machine that converts kinetic energy of the fast moving winds into electrical energy. On the basis of axis of rotation of the blades, it is divided into two parts.

1. Horizontal axis wind turbine (HAWT)

2. Vertical axis wind turbine (VAWT)



1. Horizontal Axis Wind Turbine (HAWT)

It is a turbine in which the axis of rotation of rotor is parallel to the ground and also parallel to wind direction.

They are further divided into two types

(i) Upwind turbine

(ii) Downwind turbine

(i) Upwind Turbine

The turbine in which the rotor faces the wind first are called upwind turbine.

- Today most of the HAWT is manufactured with this design.
- This turbine must be inflexible and placed at some distance from the tower.
- The basic advantage of this turbine is that, it is capable of avoiding wind shade behind the tower.
- It requires yaw mechanism, so that its rotor always faces the wind.

(ii) Downwind Turbine

The turbine in which the rotor is present at the downside of the tower is called downwind turbine. In these types of wind turbines, the wind first faces the tower and after that it faces the rotor blades.

- Yaw mechanism is absent in this turbine. The rotors and nacelles are designed in such a way that the nacelle allows the wind to flow in a controlled manner.

- It receives some fluctuation in wind power because here the rotor passes through the wind shade of the tower. In other words the rotor is present after nacelle of the tower and this create fluctuation in the wind power.

Advantages and Disadvantages of HAWTs

The various advantages and disadvantages of the horizontal axis types of wind turbines are:

Advantages

- It has self-starting ability. It does not require any external power source to start.
- It has high efficiency as compared with the HAWT.
- Capable of working in high wind speed condition.
- In the case of slow wind condition, its angle of attack can be varied to get maximum possible efficiency.
- Since all blades of this turbine work simultaneously, so it is capable of extracting maximum energy form the wind.

Disadvantages

- Its initial installation cost is high.
- It requires large ground area for its installation.
- Because of its giant size of blades and towers, it becomes difficult to transport it to the sites.
- High maintenance cost.
- Creates noise problem.
- It cannot be installed near human population.
- It is not good for the bird's population. They are killed by its blades rotation.

2. Vertical Axis Wind Turbine (VAWT)

It is a turbine in which the axis of rotation of the rotor is perpendicular to the ground and also perpendicular to the wind direction.

- It can operates in low wind situation.
- It is easier to build and transport.
- These types of Wind turbines are mounted close to the ground and are capable of handling turbulence in far better way as compared with the HAWT.
- Because of its less efficiency, it is used only for the private purpose.

VAWTs are further classified as

(i) Darrieus turbine

(ii). Giromill turbine

(iii) Savonius turbine

(i) Darrieus Turbine

Darrieus turbine is type of HAWT. It was first discovered and patented in 1931 by French aeronautical engineer, Georges Jean Marie Darrieus. It is also known as egg beater turbine because of its egg beater shaped rotor blades.

- It consists of vertically oriented blades which are mounted on a vertical rotor. It is not a self-starting turbine and hence a small powered motor is required to start its rotation.

- First the Darrieus turbine is rotated by using a small powered motor. Once it attains sufficient speed, the wind flowing across its blades generates lift forces and this lift forces provides the necessary torque for the rotation. As the rotor rotates, it also rotates the generator and electricity is produced.

(ii) Giromill Turbine:

It is similar to the Darrieus turbine but the difference is that, it has H-shaped rotor. It works on the same principle of Darrieus turbine.

- This turbine has H- shaped rotor. Here Darrieus design which has egg beater shaped rotor blades are replaced by straight vertical blades attached with central tower with horizontal supports. It may consists of 2-3 rotor blades.
- Giromill turbine is cheap and easy to build as compared with Darrieus turbine. It is less efficient turbine and requires strong wind to start. Same as darrieus types of wind turbines, it is also not self- starting and requires small powered motor to start. It is capable of working in turbulent wind conditions.

(iii) Savonius Turbine

Savonius turbine is HAWT.It was first discovered in 1922 by a Finnish Engineer Sigurd Johannes Savonius. It is one of the simplest turbine among all known turbines.

- It is a drag-type device and consists of two or three scoops. If we look it from above than it looks 'S' shape in cross section. The scoops of these turbines have curvature shape and because of that, it experiences less drag when it moves against the wind instead of moving with the wind.
- Since it is a drag-type machine, it is capable of extracting very less amount of wind power as compared with other similar sized lift-type turbines.

Advantages and Disadvantages of Vertical Axis Wind Turbine

Advantages

- It is simple in design and easy to construct and transport.
- It can be easily installed to desired location.
- It requires less ground area for its installation.
- Initial installation cost is very less as compared with the HAWT.
- It can work in turbulent wind condition.
- It is omni-directional and hence do not need to track winds.
- They are smaller in size and hence can be used for domestic or private purpose easily.
- They have low maintenance cost as compared with the HAWT.

Disadvantages

- It is less efficient. The efficiency of this turbine is about 30-35%.
- They are not self-starting. A small powered motor is needed to start it.
- Guy wires may required to support this turbine.

Difference between Horizontal Axis Wind Turbine and Vertical Axis Wind Turbine

The various difference between horizontal axis and vertical axis types of wind turbines in tabular form are given below:

S.no	Horizontal Axis Wind Turbine	Vertical Axis Wind Turbine
1.	In HAWTs, the axis of rotation of the rotor is Horizontal to the ground.	In VAWTs the axis of rotation of the rotor is perpendicular to the ground.
2.	Yaw mechanism is present.	Absence of Yaw mechanism.
3.	It has high initial installation cost.	It has low initial installation cost.
4.	They are big in size.	They are small in size.
5.	Its efficiency is high.	It has low efficiency.
6.	It requires large ground area for installation.	It requires less ground area for installation.
7.	High maintenance cost.	Low maintenance cost as compared with HAWT.
8.	They are self-starting.	They are not self-starting.
9.	They are unable to work in low wind speed condition.	They are capable of working in low wind speed condition.
10.	Difficult in transportation.	Easy in transportation.
11.	They are mostly used commercially.	They are mostly used for private purpose only.
12.	It cannot be installed near human population.	It can be installed near human population.
13.	It is not good for the bird's population.	It is good for the bird's population.

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6. Explain the components of wind electric system with neat sketches?

6.11.1. Introduction

The basic components of a wind-electric conversion system are shown in Fig. (6.11.1). Aeroturbines convert wind energy into rotary mechanical energy. A mechanical interface, consisting of a step-up gear

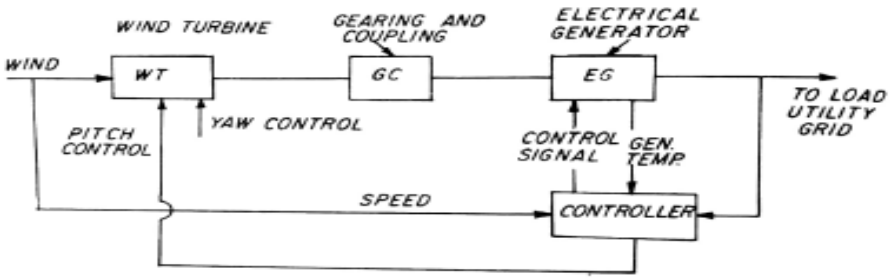


Fig. 6.11.1. Components of wind electric system.

and a suitable coupling transmits the energy to an electrical generator. The output of this generator is connected to the load or system grid. The controller senses the wind direction, wind speed, power output of the generator and other necessary performance quantities of the system and initiates appropriate control signals to take suitable corrective actions. The system should be protected from excessive temperature rise of the generator, electrical faults and extra wind conditions.

The choice of an electrical generator and control method to be employed (if any) can be decided by consideration of the following three factors :

(*i*) the basis of operation *i.e.* either constant tip speed or constant tip speed ratio

(*ii*) the wind-power rating of the turbine and

(*iii*) the type of load demand *e.g.* battery connection.

Wind power ratings can be divided into three convenient grouping, *small* to 1 kW, *medium* to 50 kW and *large* 200 kW to megawatt frame size.

Electrical generators types applicable to each of these ratings are :

Small—permanent, magnet, d.c. generators.

Medium—permanent magnet, d.c. generator, induction generator, synchronous generator.

Large—induction generator, synchronous generator.

The electrical control strategy employed for any particular scheme can be designed to effect control of the generator, the power transmission link or the load.

7. a. Explain the schemes of electric generation in WECS?

6.11.2. Schemes for electric generation

Several schemes for electric generation have been developed. These schemes can be broadly classified under three categories :

(*i*) Constant—speed constant frequency systems (CSCF)

(*ii*) Variable speed constant frequency systems (VSCF)

and (*iii*) Variable speed variable frequency systems (VSVF).

(1) **Constant speed constant frequency system (CSCF).** Constant speed drive has been used for large generators connected directly to the grid where constant frequency operation is essential.

(*ii*) **Variable speed constant frequency scheme.** (VSCF Scheme). Variable-speed drive is typical for most small wind generators used in autonomous applications, generally producing variable frequency and variable voltage output. The variable speed operation of wind-electric system yield higher outputs for both low and high wind speeds. This results in higher annual energy yields per rated installed kW capacity. Both horizontal axis and vertical axis turbines will exhibit this gain under variable speed operation. The popular schemes to obtain constant frequency output are as follows :

(*iii*) **Variable Speed Variable Frequency (VSVF) Schemes.** Since resistive heating loads are essentially frequency insensitive, the

a.c. generator can be effected at a variable frequency corresponding t the changing derive speed. For this purpose capacitor excited (self-ex cited) squirrel caze induction machines can be conveniently used. Such a scheme is shown in Fig. (6.11.5). These systems are gaining importance for stand alone wind power applications. The magnitude and frequency of the emf depends on the value of the load impedance, prime-mover speed and excitation capacitance. Methods of analysing these variable—voltage variable frequency generators have been developed to predict the no-load and load performance characteristics. Computer programmes have also been formulated for this purpose.

The varying output voltage can be converted to constant d.c. using choppers or controlled rectifiers on constant a.c. using force—commutated inverters. AC converters and transducers can be introduced to monitor and control the desired performance quantities.

b. Explain the environmental aspects of WECS?

Wind power plant as a renewable energy source has slight influence on environment, nevertheless people reluctantly agree to construct power plants in their neighbourhood.

One of the most common environmental aspect regard noise. The noise has two sources: mechanical (inherent in the gearing system – excluding double feed system) and the second one related with aerodynamics of the rotor blade. The first one is possible to eliminate but the second one not. The aerodynamic noise arise when the rotating blade passing the tower. That effect calls the tower thumb. The most influence on the level of loud has wind speed. When the wind blow fast the background noise is enough loud to drown out the tower thumb effect but when the wind is blowing lightly the tower thumb effect is audible in long distance. [3]

Other environmental problem is electromagnetic interference. Large structures like wind rotor and tower can cause objectionable electromagnetic interference in the performing of a nearby transmitters or receivers. Moreover the rotating blades can also reflect signals what make experience interference at the blade passage frequency. The highest influence on that effect strength has location. This problem is important for onshore technology but in some of case large offshore power plants the problem could be higher. The offshore power plant can interference with radar and flight paths to airfields.

Next one environmental problem is effect on birds. The wind turbine blade is a lethal weapon against any avian population. The birds may be killed or at least injured if the collide with blade. The most often situation the suction draught created by wind flowing to a turbine caught the birds and poke them into air stream headed for the blades. Extremely dangerous are turbines with lattice towers, where the bird has possibility to nest. Therefore often wind farms have to be sited away avian flight paths.

The last one environmental problem is visual impact mainly for onshore wind farms. The problem is related with property owners around the wind farm, who do not allow for wind power plant installation in their neighbourhood. However that problem also regard offshore technology for farms installed near resorts.

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